Lab 8

ENGI25469

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LAB REPORT

Introduction:

The lab focused on enhancing our robotic programming abilties through enforcing timers, counters, offsets, and conditional logic. Building upon preceding labs, we modified an current application to encompass these superior capabilities, aiming to create a greater complex and dynamic robot operation. The primary targets had been to implement a timer for cycle time measurement, create a counter for monitoring application iterations, follow dynamic offsets to robot actions, and comprise conditional logic for application waft manage.

Aim:

The aim of this lab is to decorate our robotic programming capabilities by using implementing timers, counters, offsets, and conditional good judgment to create a extra dynamic and complex robotic operation. Specifically, the lab focuses on measuring cycle times, counting application iterations, applying dynamic offsets to robotic actions, and the usage of conditional branches to control application float.

Procedure (Screenshot are attached at the end):

Part 1: Timer Implementation

1. Created a copy of the Lab 5 Shapes program, naming it Final_[JKHA].
2. Added instructions to reset and start a Timer at the beginning of the program.
3. Created a data Register named Time_[JKHA].
4. Added an instruction to stop the Timer and store its value in the Time Register at the end of the program.

Part 2: Counter Implementation

5. Initialized a counter named Cycle_Count_[JKHA] at the beginning of the program, setting it to 0.
6. Added an instruction to increment the counter after each completion of the Semi-Circle movement.

Part 3: Offset Implementation

1. Created a Position Register named Offset with X, Y, Z, W, P, and R values set to 0.
2. Added an instruction to initialize the Z value of the Offset Position Register to 0 at the start of the program.
3. Implemented logic to increment the Z element of the Offset Position Register by 15 mm after each Semi-Circle movement completion.

Part 4: Logic Implementation

1. Created a conditional branch after the Counter and Offset increments.
2. Programmed the branch to jump to the start of the Square positions if the Cycle Count is less than 3.
3. Modified Square and Semi-Circle positions to incorporate the Offset Position Register values.

Part 5: Testing

4. Tested the program in Step mode, then in Continuous mode.
5. Verified that the robot runs 3 cycles, with cycles 2 and 3 rising 15 mm from the previous path.
6. Recorded a video showing the teach pendant program and the robot running in continuous mode.
7. Captured a screenshot of the Time Register after testing to show the total cycle time.

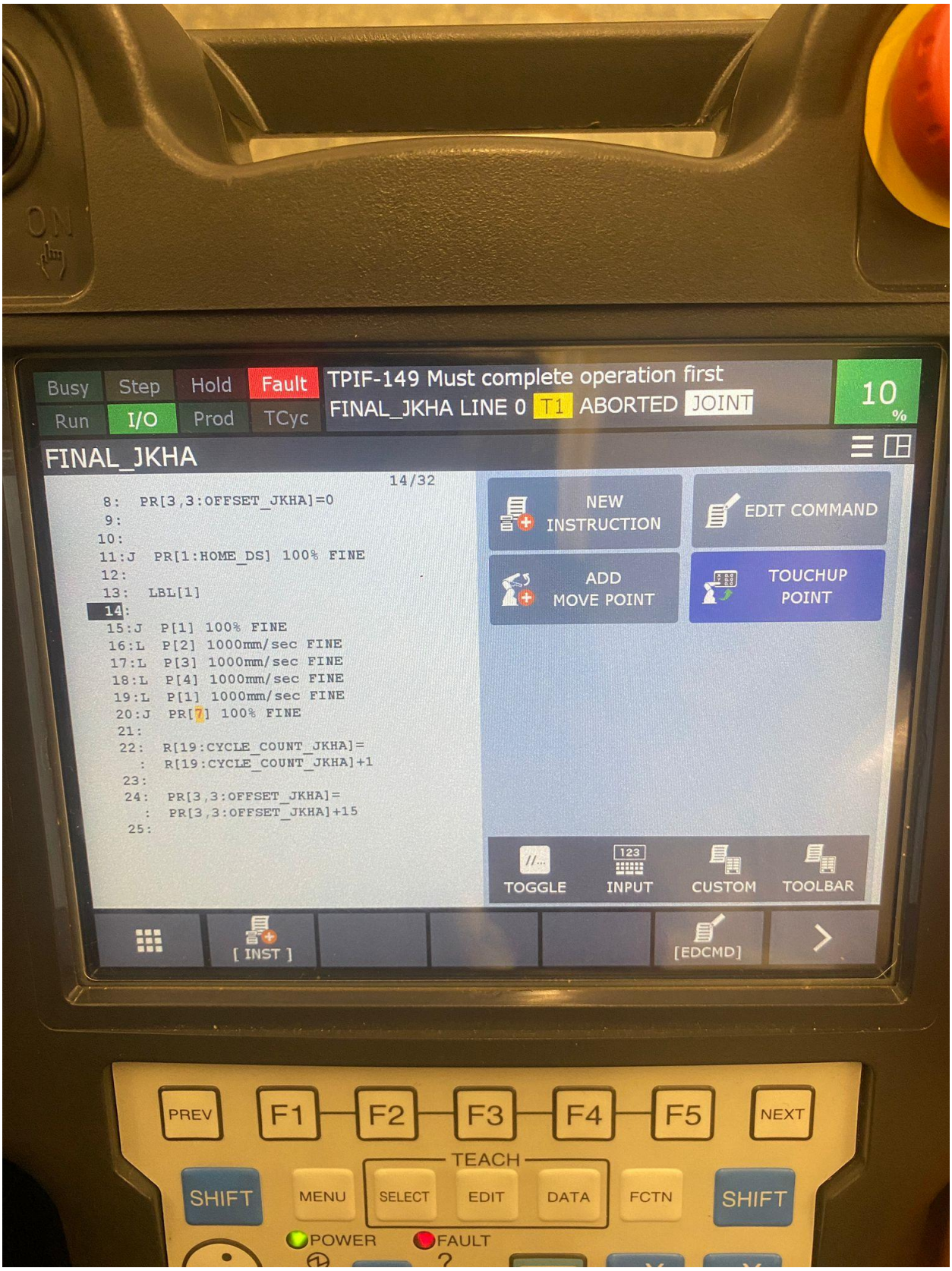
Conclusion:

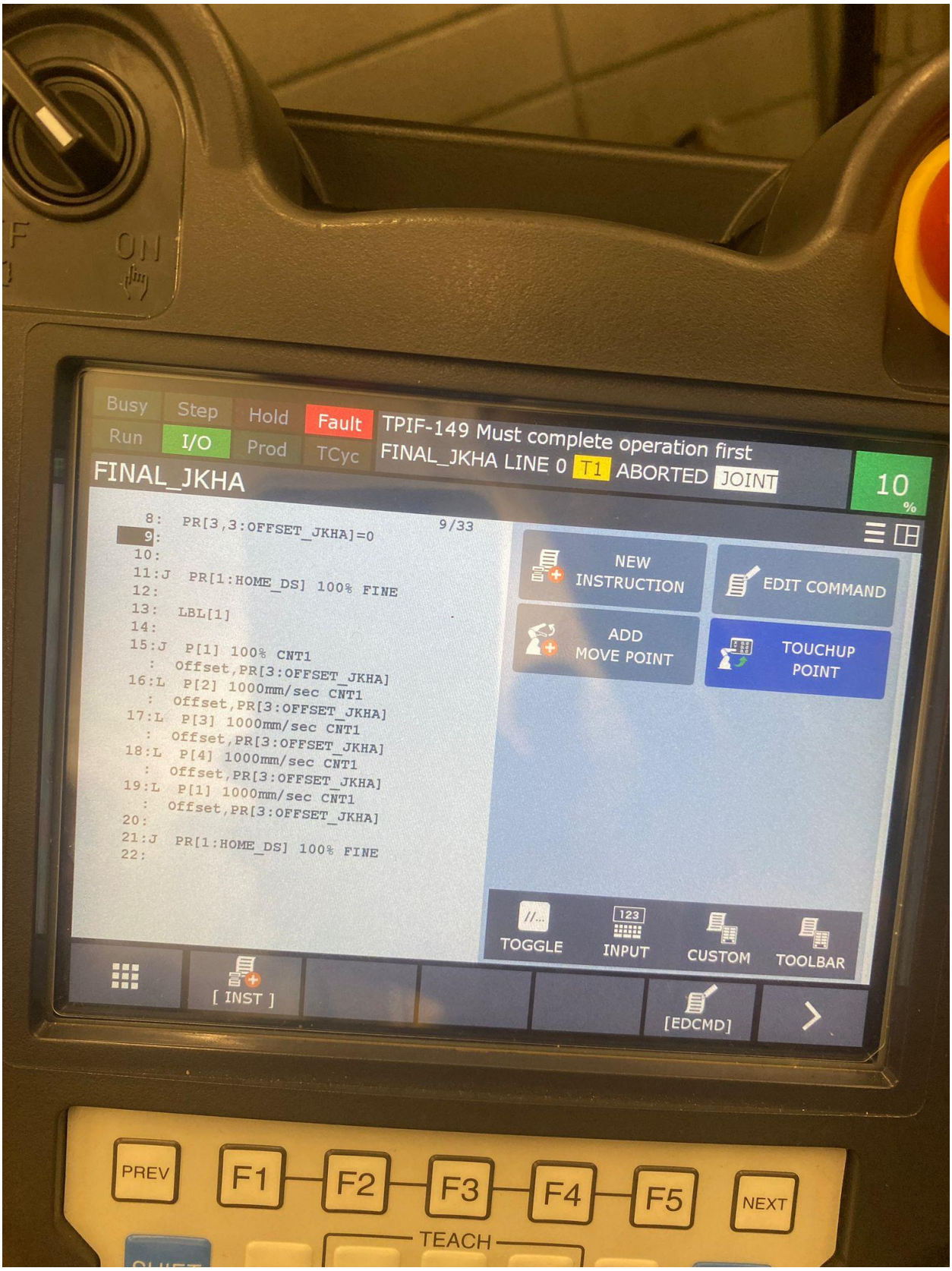
This lab effectively improved our robot programming abilities with the aid of incorporating advanced ideas along with timers, counters, offsets, and conditional good judgment. The implementation of those functions allows for more complex and flexible robot operations, essential in business packages. The potential to music cycle instances, rely iterations, and dynamically alter robot moves demonstrates the flexibility and precision attainable in modern-day robot

structures. These skills are essential for developing green and adaptable robot applications in real-global business settings.

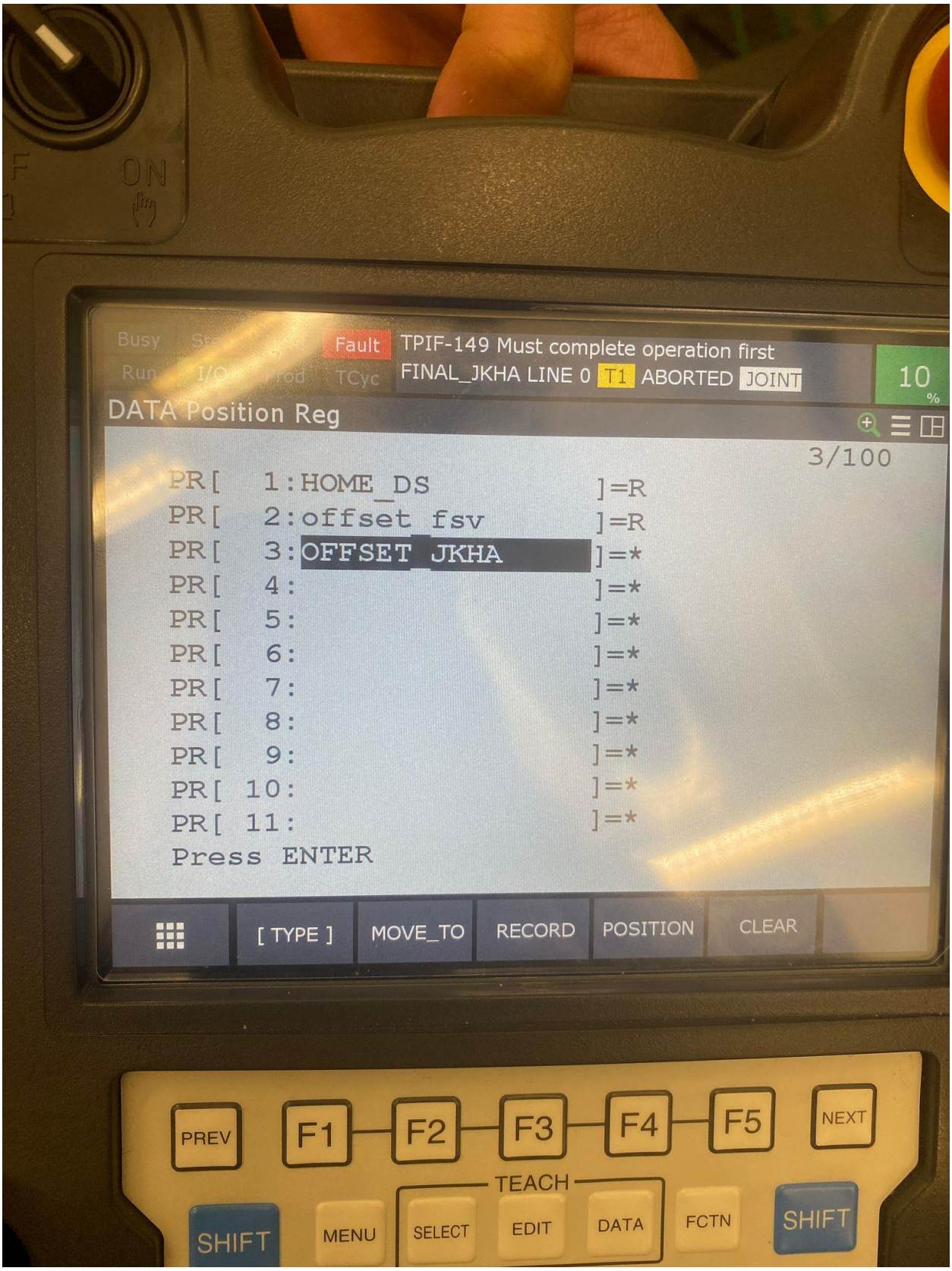
Screenshots:

LAB REPORT


Busy
Step
Hold
Fault
Run
I/O
Prod
TCyc
TPIF-149 Must complete operation first
FINAL_JKHA LINE 0 T1 ABORTED JOINT
10 %
FINAL_JKHA
14/32
8: PR[3,3:OFFSET_JKHA]=0
9:
10:
11:J PR[1:HOME_DS] 100% FINE
12:
13: LBL[1]
14:
15:J P[1] 100% FINE
16:L P[2] 1000mm/sec FINE
17:L P[3] 1000mm/sec FINE
18:L P[4] 1000mm/sec FINE
19:L P[1] 1000mm/sec FINE
20:J PR[7] 100% FINE
21:
22: R[19:CYCLE_COUNT_JKHA]=
: R[19:CYCLE_COUNT_JKHA]+1
23:
24: PR[3,3:OFFSET_JKHA]=
: PR[3,3:OFFSET_JKHA]+15
25:
NEW INSTRUCTION
EDIT COMMAND
ADD MOVE POINT
TOUCHUP POINT
TOGGLE
INPUT
CUSTOM
TOOLBAR
[INST]
[EDCMD]
PREV
F1
F2
F3
F4
F5
NEXT
TEACH
SHIFT
MENU
SELECT
EDIT
DATA
FCTN
SHIFT
POWER
FAULT


Busy
Step
Hold
Fault
Run
I/O
Prod
TCyc
TPIF-149 Must complete operation first
FINAL_JKHA LINE 0 T1 ABORTED JOINT
FINAL_JKHA
10 %
9/33
8: PR[3,3:OFFSET_JKHA]=0
9:
10:
11:J PR[1:HOME_DS] 100% FINE
12:
13: LBL[1]
14:
15:J P[1] 100% CNT1
: Offset,PR[3:OFFSET_JKHA]
16:L P[2] 1000mm/sec CNT1
: Offset,PR[3:OFFSET_JKHA]
17:L P[3] 1000mm/sec CNT1
: Offset,PR[3:OFFSET_JKHA]
18:L P[4] 1000mm/sec CNT1
: Offset,PR[3:OFFSET_JKHA]
19:L P[1] 1000mm/sec CNT1
: Offset,PR[3:OFFSET_JKHA]
20:
21:J PR[1:HOME_DS] 100% FINE
22:
NEW INSTRUCTION
EDIT COMMAND
ADD MOVE POINT
TOUCHUP POINT
TOGGLE
INPUT
CUSTOM
TOOLBAR
[INST]
[EDCMD]
PREV
F1
F2
F3
F4
F5
NEXT
TEACH

LAB REPORT


Busy
Run
Fault
TPIF-149 Must complete operation first
FINAL_JKHA LINE 0 T1 ABORTED JOINT
10 %
DATA Position Reg
3/100
PR[1:HOME_DS]=R
PR[2:offset_fsv]=R
PR[3:OFFSET_JKHA]=*
PR[4:]=*
PR[5:]=*
PR[6:]=*
PR[7:]=*
PR[8:]=*
PR[9:]=*
PR[10:]=*
PR[11:]=*
Press ENTER
[TYPE]
MOVE_TO
RECORD
POSITION
CLEAR
PREV
F1
F2
F3
F4
F5
NEXT
TEACH
SHIFT
MENU
SELECT
EDIT
DATA
FCTN
SHIFT